

分治法

《算法设计与分析》

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04



归 并 排 序



归并排序问题

- 该问题是将 n 个元素排成非递减顺序。

- 算法描述

若 n 为 1，算法终止；否则，将这一元素集合分割成两个或更多个子集合，对每一个子集合分别排序，然后将排好序的子集合归并为一个集合。

MERGE-SORT(A, p, r)

if $p < r$ **then** $q \leftarrow (p + r)/2$

MERGE-SORT(A, p, q)

MERGE-SORT($A, q + 1, r$)

MERGE(A, p, q, r)

MERGE(A, p, q, r)

$n_1 \leftarrow q - p + 1$; $n_2 \leftarrow r - q$

create arrays $L[1.. n_1 + 1]$ and $R[1.. n_2 + 1]$

for $i \leftarrow 1$ **to** n_1 **do** $L[i] \leftarrow A[p + i - 1]$

for $j \leftarrow 1$ **to** n_2 **do** $R[j] \leftarrow A[q + j]$

$L[n_1 + 1] \leftarrow \infty$; $R[n_2 + 1] \leftarrow \infty$

$i \leftarrow 1$; $j \leftarrow 1$

for $k \leftarrow p$ **to** r

do if $L[i] \leq R[j]$

then $A[k] \leftarrow L[i]$

$i \leftarrow i + 1$

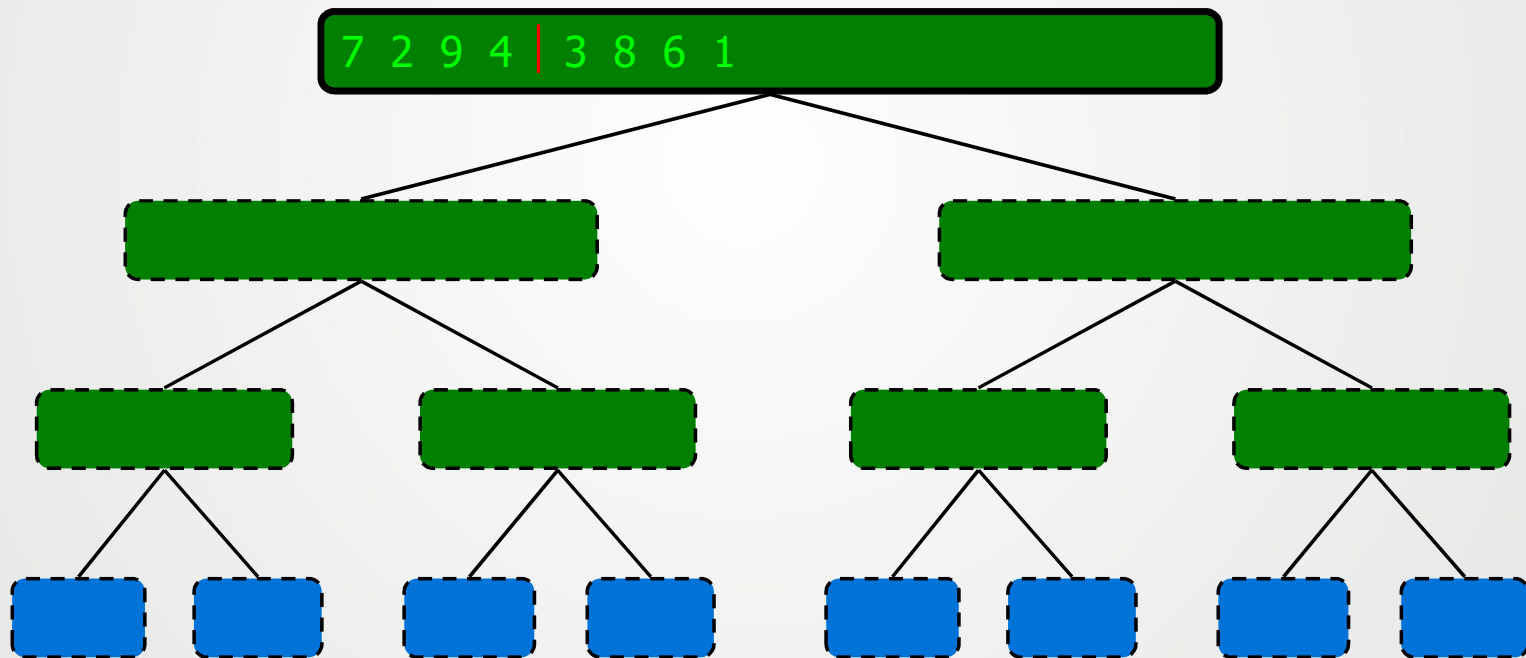
else $A[k] \leftarrow R[j]$

$j \leftarrow j + 1$

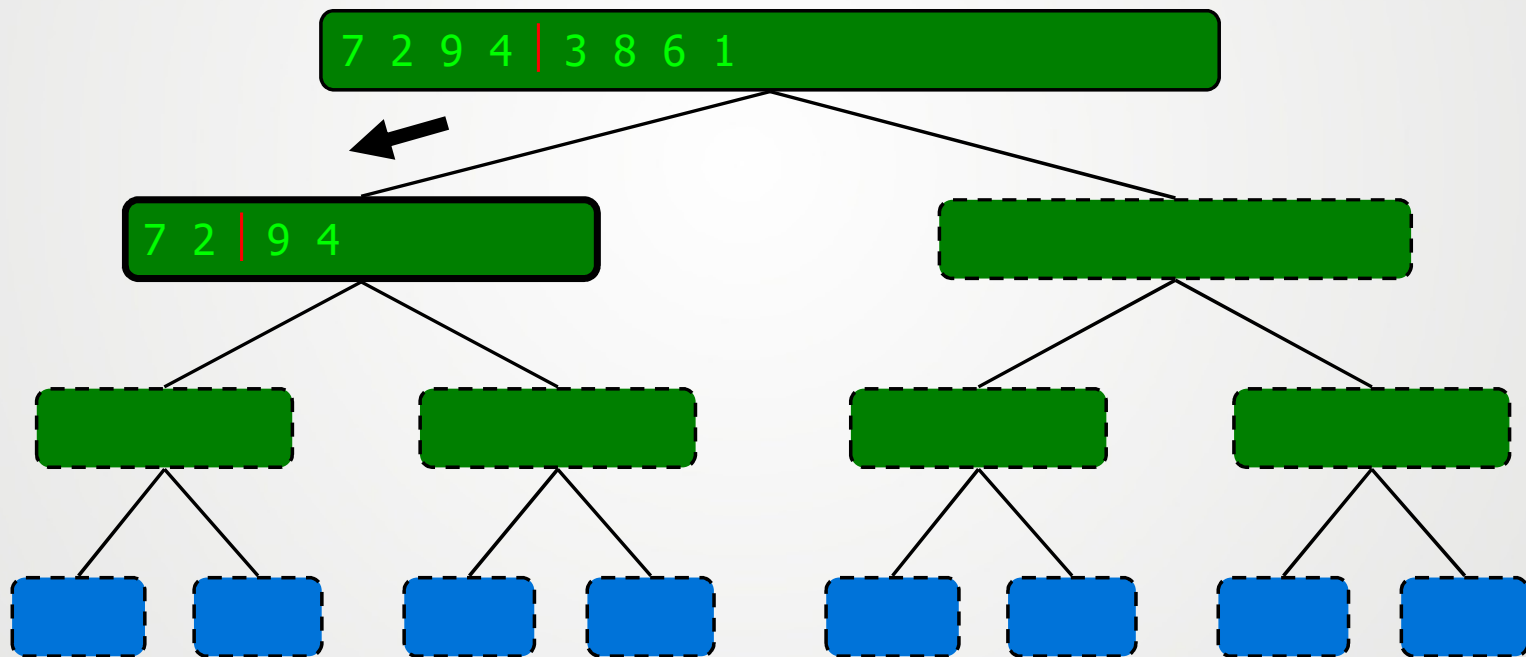


Execution Example

- Partition

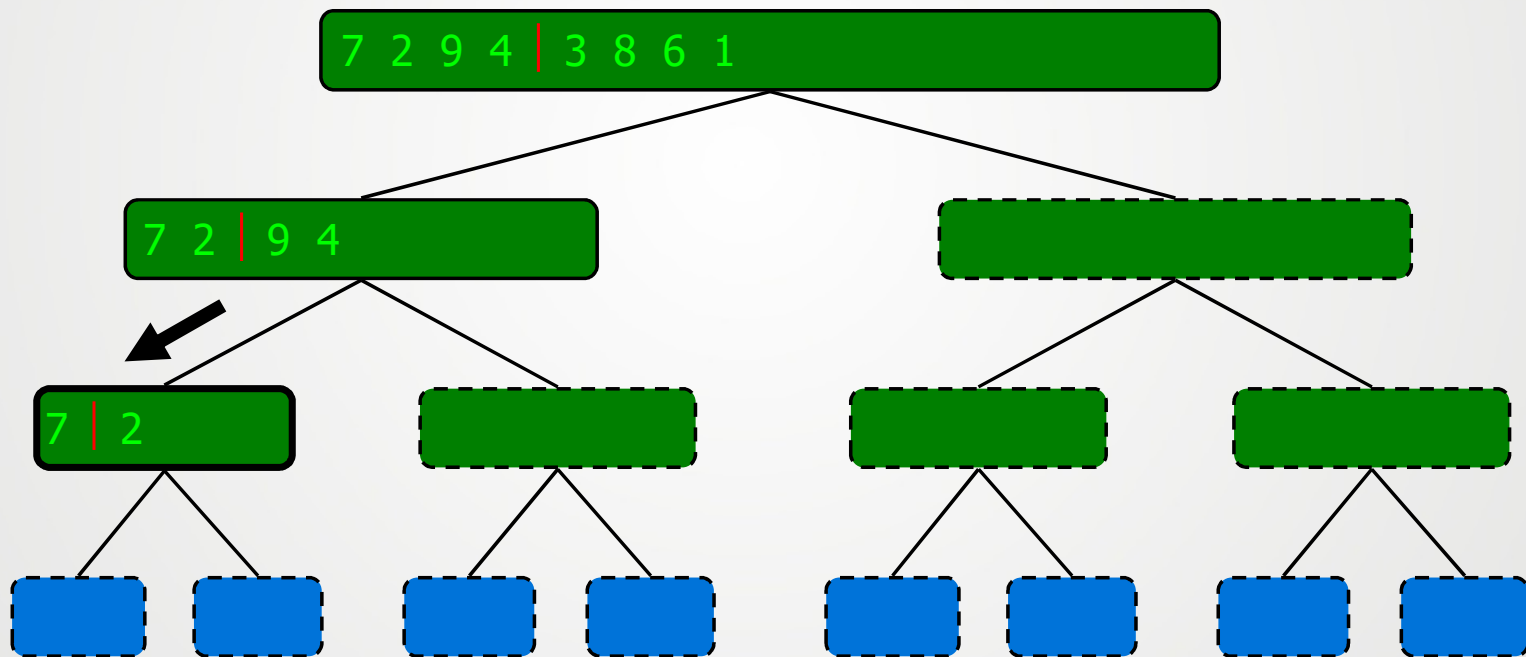


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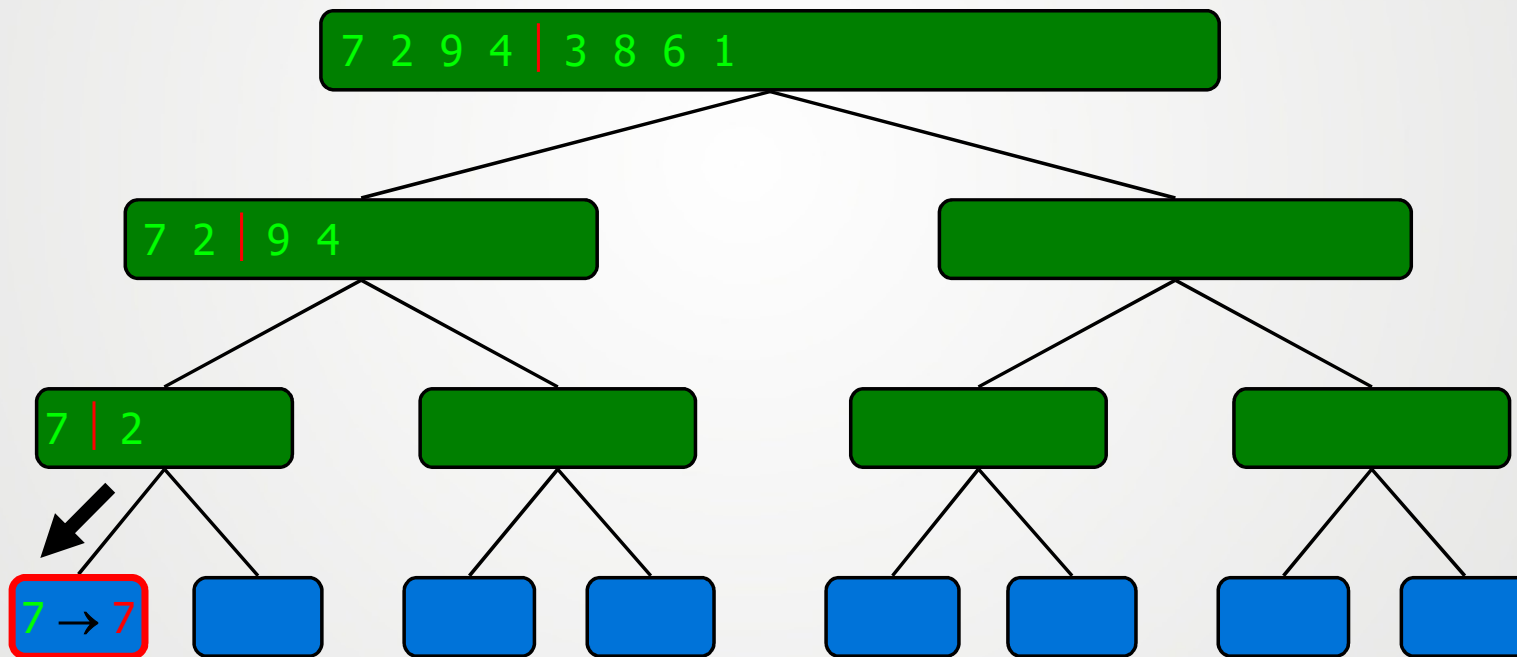
>>> Execution Example (cont.)

- Recursive call, partition



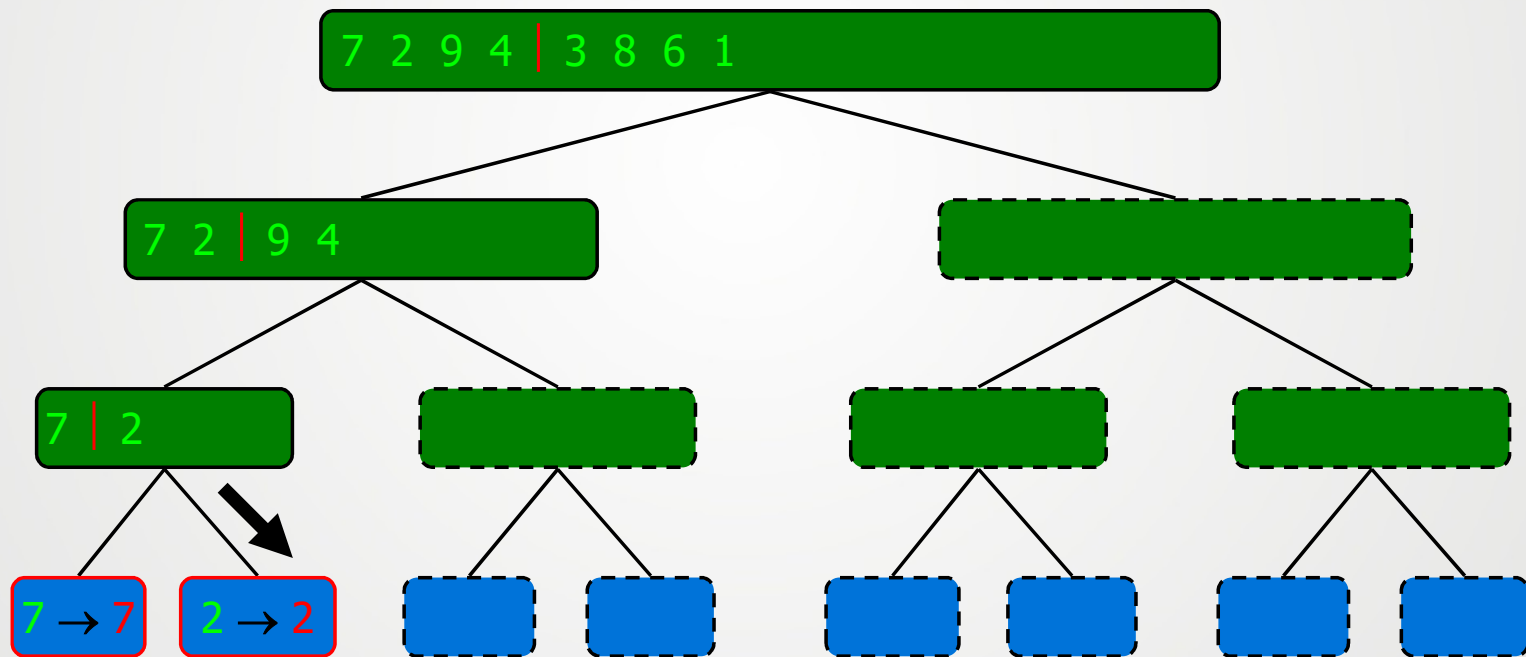
>>> Execution Example (cont.)

- Recursive call, base case



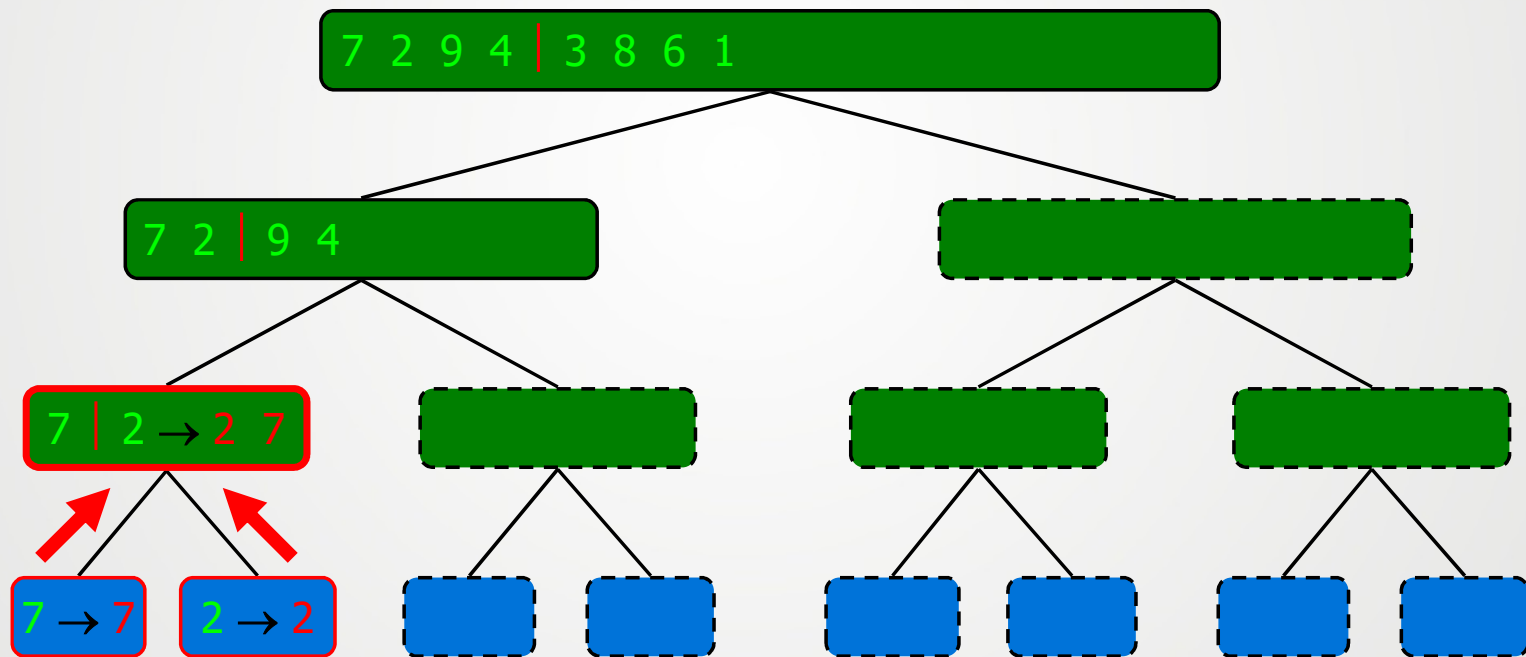
>>> Execution Example (cont.)

- Recursive call, base case



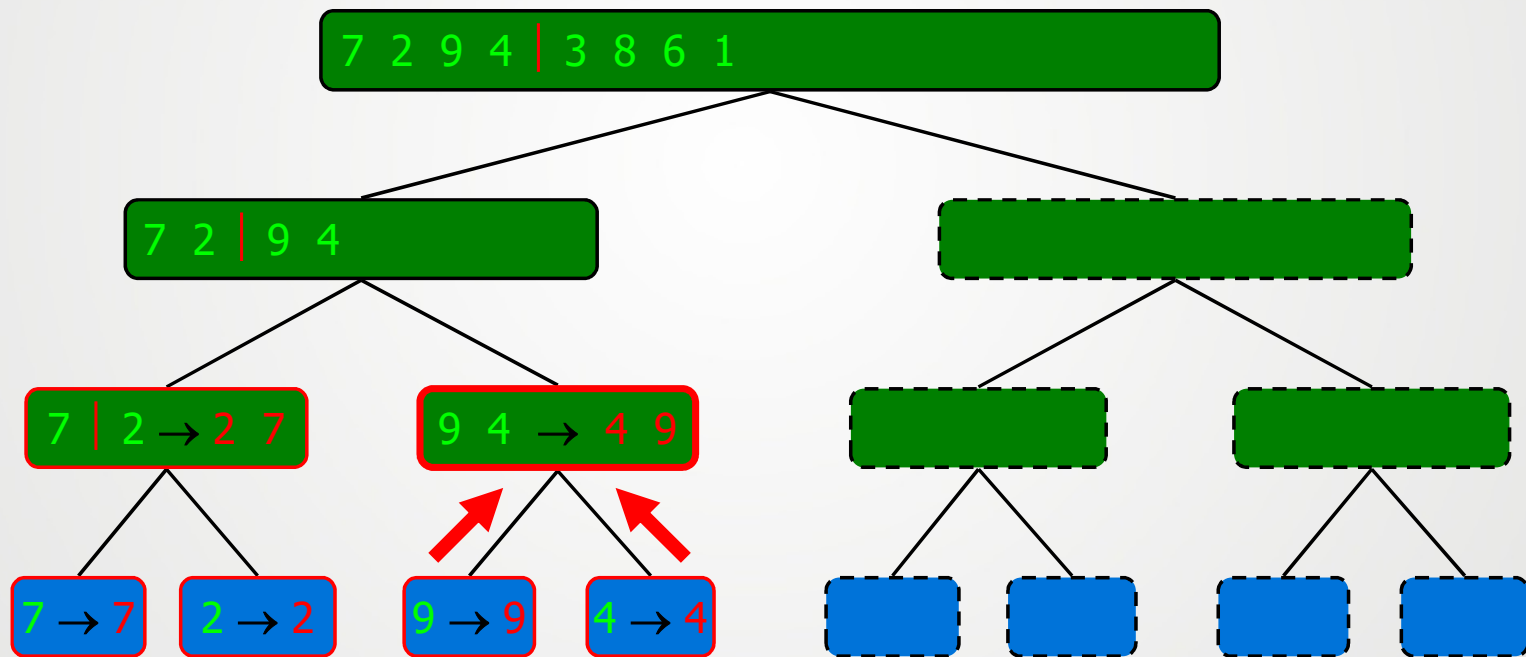
>>> Execution Example (cont.)

- Merge



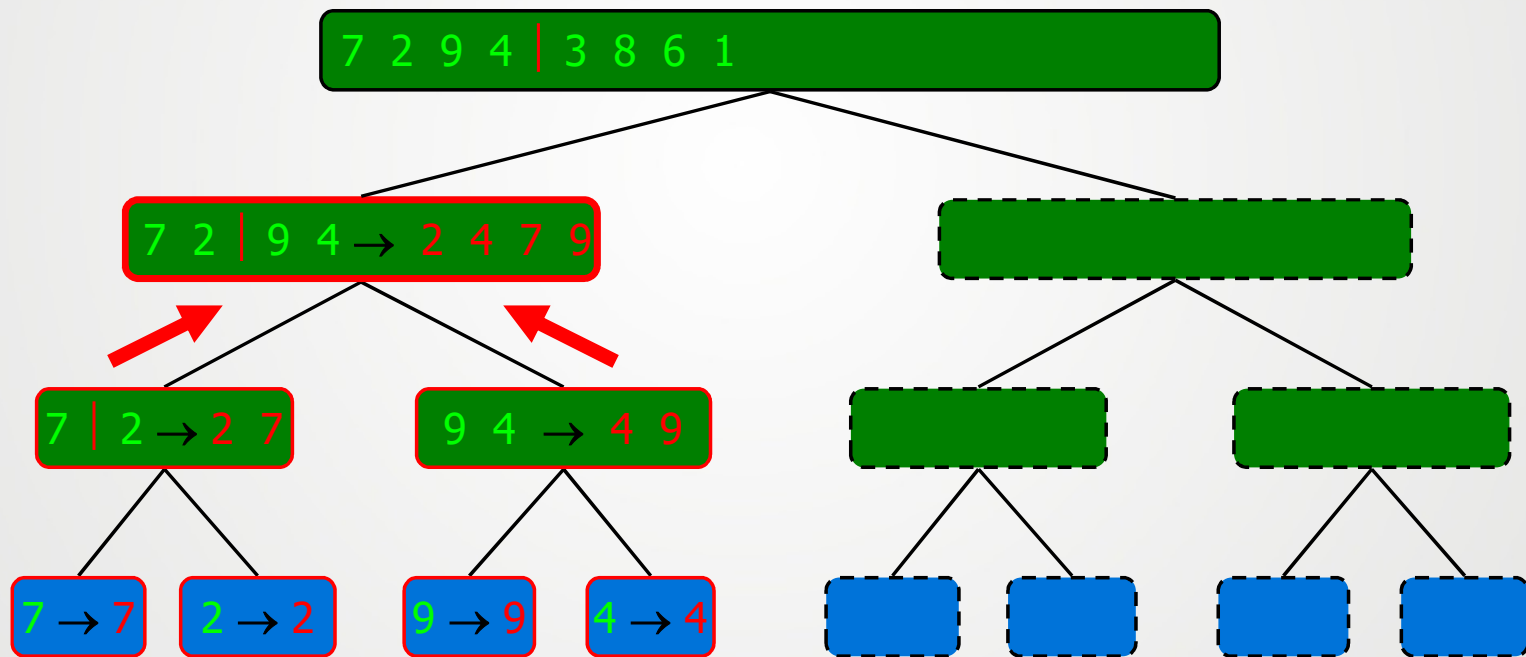
>>> Execution Example (cont.)

- Recursive call, ..., base case, merge



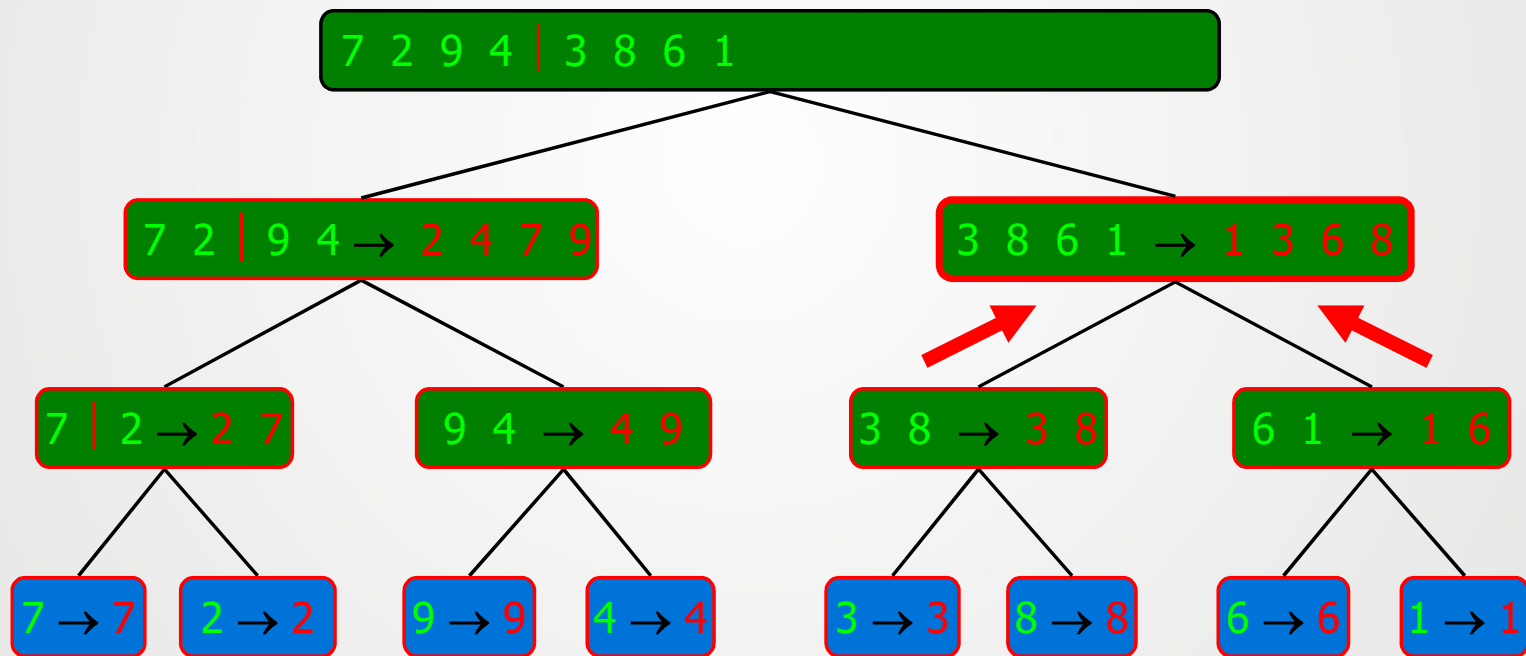
Execution Example (cont.)

- Merge



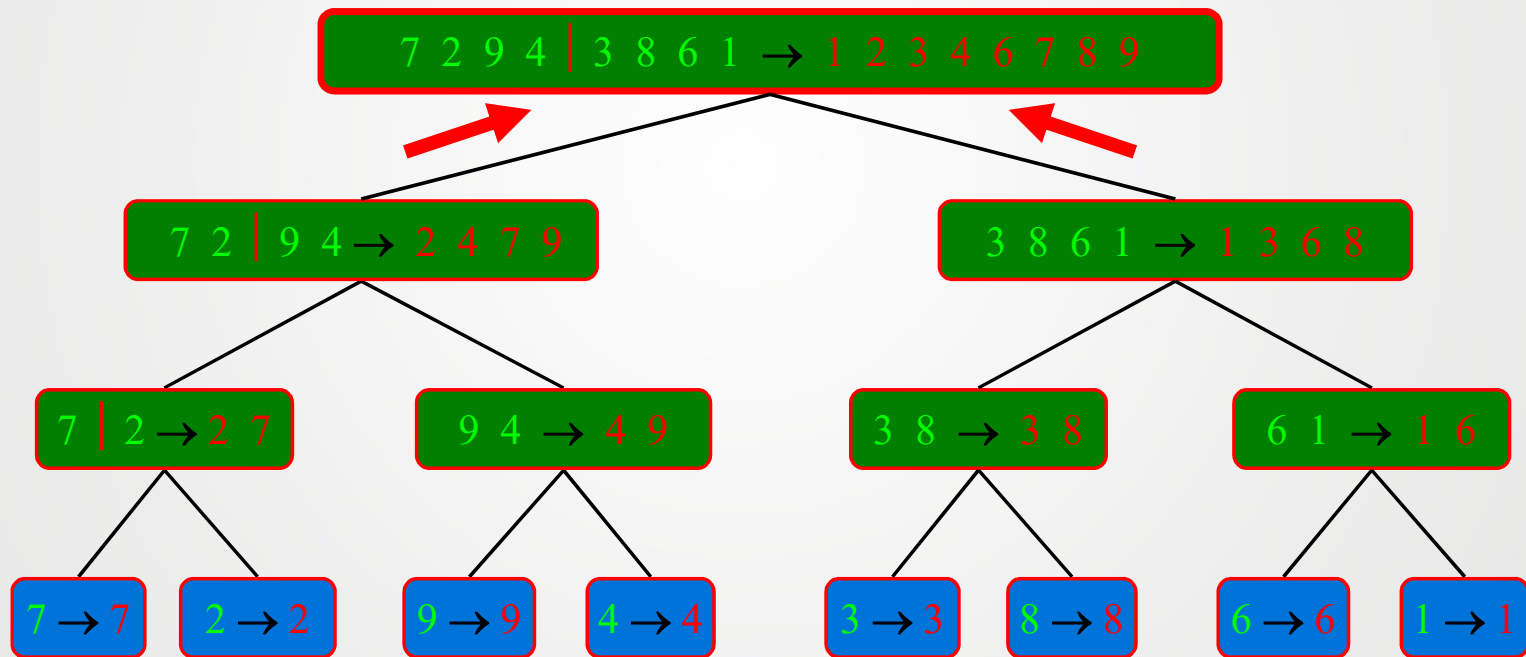
>>> Execution Example (cont.)

- Recursive call, ..., merge, merge



Execution Example (cont.)

■ Merge





归并排序

 最坏时间复杂度: $O(n \log n)$

 平均时间复杂度: $O(n \log n)$

 辅助空间: $O(n)$



归并排序

算法mergeSort的递归过程可以消去。

初始序列

[49] [38] [65] [97] [76] [13] [27]

第一步

[38 49] [65 97] [13 76] [27]

第二步

[38 49 65 97] [13 27 76]

第三步

[13 27 38 49 65 76 97]

A hand is shown drawing a circle on a whiteboard with a black marker. To the left, a document with a colorful circular diagram is visible. The background is a bright, modern office space with large windows. The text '谢谢大家' is overlaid in a large, bold, blue font on the right side of the image.

谢谢大家